



University
Mohammed VI
Polytechnic

Quadcopter Autonomous Inspection Internship

From flight simulation to AI-assisted inspection

A two-week engineering experience integrating **PX4 Autopilot**, **ROS 2 Humble**, **Gazebo**, trajectory generation, onboard sensing, and **YOLOv8** perception.

ORGANIZED BY

School of Applied Sciences & Engineering

Vanguard Center | Mohammed VI Polytechnic University

Track: Autonomous Drone Engineering

Format: Full-time, team-oriented, learning by doing



VANGUARD CENTER

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Welcome to the internship. This document presents the mission, engineering architecture, learning objectives, working method, and first onboarding steps. The repository `README.md` inside `quad_um6p_internship` project directory remains the authoritative operational starting point.

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1 Project Overview

The **Quadcopter Autonomous Inspection Internship** introduces the complete chain of a modern autonomous aerial system. The project combines a simulated quadcopter, the PX4 flight stack, ROS 2 communication, Gazebo-based experimentation, onboard camera data, and YOLOv8 object detection to perform a structured inspection mission.

The mission is easy to state but technically rich: the quadcopter takes off, follows a planned trajectory, acquires images, identifies selected objects, associates detections with flight information, and produces evidence that the mission was performed correctly.

2 Technical Architecture

Component	Role in the project
Gazebo	Simulates the world, vehicle, sensors, camera, gravity, motion, and inspection targets in a safe and repeatable laboratory.
PX4 Autopilot	Provides state estimation, flight modes, position and attitude control, safety behavior, and actuator-level control.
ROS 2 Humble	Connects modular nodes through topics, services, parameters, Quality of Service policies, and recorded data.
Micro XRCE-DDS	Bridges PX4 uORB information and ROS 2 topics so external nodes can observe the vehicle and publish mission references.
Orbit inspection node	Generates warmup, climb, approach, circular inspection, hold, and landing phases.
YOLOv8 perception	Processes camera images, identifies selected classes, annotates images, and produces measurable detection evidence.

Mission → Requirements → Architecture → Design → Simulation
Programming → Integration → Testing → Evidence → Demonstration

3 Goals and Objectives

3.1 Technical objectives

By the end of the internship, students should be able to:

- explain the architecture of an autonomous quadcopter system;
- distinguish the responsibilities of Gazebo, PX4, ROS 2, and YOLOv8;
- use Ubuntu terminals to inspect, launch, monitor, and stop robotics processes;
- understand ROS 2 nodes, topics, services, parameters, launch files, and QoS;

- interpret PX4 local North-East-Down coordinates;
- understand how position and yaw references form a trajectory;
- run and modify a reproducible Gazebo inspection mission;
- bridge simulated camera images into ROS 2;
- interpret YOLOv8 classes, confidence values, bounding boxes, missed detections, and false detections;
- collect evidence using rosbags, logs, plots, and reports;
- compare desired and measured behavior with meaningful metrics;
- combine individual contributions into one coherent team demonstration.

3.2 Educational objectives

- decompose an open problem into testable subsystems;
- make assumptions explicit and verify them experimentally;
- change one factor at a time when diagnosing behavior;
- treat failure as information when it is observed and understood;
- separate configuration, mathematics, control logic, monitoring, and evidence;
- write readable code and reproducible procedures;
- communicate technical choices and limitations honestly;
- collaborate through clear interfaces and individual accountability.

4 Engineering and Physics Foundations

4.1 Coordinate frames

PX4 local position uses North-East-Down:

$$\mathbf{p}_{\text{NED}} = \begin{bmatrix} N & E & D \end{bmatrix}^T.$$

Positive vertical position points downward, so altitude h above the local origin is commanded as $D = -h$. Visualization commonly uses East-North-Up:

$$(x_{\text{ENU}}, y_{\text{ENU}}, z_{\text{ENU}}) = (E, N, -D).$$

A numerically correct value in the wrong frame remains an incorrect engineering command.

4.2 Circular inspection trajectory

A horizontal orbit of radius R , centered at (N_c, E_c) , is

$$N(t) = N_c + R \cos \theta(t), \quad E(t) = E_c + R \sin \theta(t), \quad D(t) = -h,$$

with

$$\theta(t) = \frac{2\pi t}{T}.$$

The tangential speed and centripetal acceleration are

$$v = \frac{2\pi R}{T}, \quad a_c = \frac{v^2}{R} = \frac{4\pi^2 R}{T^2}.$$

Halving the orbit period doubles the requested speed and multiplies the required centripetal acceleration by four.

4.3 Smooth reference shaping

The project uses

$$s(u) = 3u^2 - 2u^3, \quad 0 \leq u \leq 1,$$

because

$$s(0) = 0, \quad s(1) = 1, \quad s'(0) = 0, \quad s'(1) = 0.$$

The reference starts and ends with zero slope. PX4 remains responsible for feedback control.

5 Two-Week Learning Path

TO DO ...

Learn > Build > Test > Fail > Understand > Improve > Demonstrate

6 Teamwork and Individual Contribution

Students work in one shared system while remaining accountable for an identifiable contribution. Contributions may cover mission requirements, trajectory analysis, ROS 2 integration, simulation design, perception, testing, data recording, or final demonstration coordination.

Each student maintains an engineering logbook that records the daily objective, learning, changes,

tests, evidence, failures, decisions, next action, and contribution to integration.

7 Testing, Evidence, and Safety

For each experiment, students follow:

**Claim → Setup → Controlled variable → Measurement → Result → Interpretation →
Next test**

The default environment is simulation. Physical flight requires separate authorization, an approved area, a responsible operator, successful prior validation, and a recovery method. Manual keyboard control and autonomous offboard control must never command the vehicle simultaneously.

8 Getting Started

The project repository is:

https://github.com/abdtmourji/quard_um6p_internship

Mandatory first action. Open the repository and read the top-level `README.md` completely. Then follow the phase-by-phase setup guides in order. Do not skip directly to the final pipeline.

9 Final Submission

TODO

10 Closing Perspective

This internship connects flight physics, control, sensing, simulation, middleware, computer vision, programming, experimentation, and technical communication through one visible mission. Students are not expected to know everything initially. Students are expected to ask precise questions, use the repository methodically, test assumptions, document evidence, and improve through repeated engineering cycles.

Understand the system. Build carefully. Test with evidence.